Advanced Databases

Assessment Brief

Item	Description
Module Title:	Advanced Databases
Module Code:	B9IS125
Module Lecturer (s):	Jagadeeswaran Thangaraj Luciana Nascimento
Programme/Cohort:	MSc in Cybersecurity MSc Information Systems with Computing
Method of Assessment:	Project/technical report
Percentage (%) Weighting:	30%
MIMLOs being assessed:	1, 2, 5
Assessment Number:	1
Individual/Group:	Group
Issue Date:	9th February 2026
Submission Date:	6th March 2026
Feedback Date:	4 weeks after submission
Feedback Strategy:	Moodle

Every student must submit their assignment on Moodle on or before the due date.

Assignments will not be accepted via Email.

Assignment Task

TASK MARK OVERVIEW		
Technical Document - 30% of Overall Mark		
	Sub Task	**Mark**
	Business Case, Scope, 7 Requirements	20
	List of Business Rules	20
	ERD to 3NF (crow's feet, datatypes) and Data Dictionary	40
	Report Structure (Table of contents, Conclusion, Bibliography)	10
	Individual Contribution	10
TOTAL MARKS		**100**

See below details of assignment brief for full information

RULES For This CA

- This CA must be 100% each student's own work.
- Students are **NOT** permitted to utilise any Generative AI tools, including GEN AI Engine in tools like Grammarly.
- Students are **NOT** permitted to use mill websites, answers from sites like Github or Previously created/submitted work.

Details of assignment brief:

Introduction:

The module's assignment requires you to demonstrate the knowledge and skills you have acquired throughout this unit by designing and creating a functional database application that addresses the tasks given below. You are required to design and implement a **hybrid database system** for an organization. The assignment is split into two (2) parts: Part A (CA1 30%) - Analysis and Design, and Part B (CA2 70%) - Implementation and Demonstration.

In order to complete this assignment, you should choose an appropriate organisation on which to conduct research and base your database application. Select an industry in which you are interested. For example, commercial airlines, commercial banking, retail, beauty, healthcare, fitness, publishing, hospitality, leisure and tourism, automotive, construction etc. From the industry selected, you should investigate the sort of data they hold and the types of transactions they carry out. For example, an educational establishment will hold data about students, staff, rooms, units, assessments and results. Their transactions might include adding students to units, allocating staff to teach units, allocating rooms to units, allocating different assessments to units and recording the results for each unit assessment the student undertakes. Do not choose an educational organisation as this has been used as an example. You should reference the business or organisation that you research.

********** **PART A – DESIGN** ********** **********

Analysis and Design -- Technical Report: (30 Marks)

The report must include the following:

a) Identify and discuss your chosen organisation, with a minimum of SIX (6) entities, and outline the likely data and information that the company would require. List and explain the functions and transactions that a database application might perform that would support some of the day to day functions of the organisation. Summarise the advantages of a DBMS to the organisation.

b) Draw an ERD (Entity Relationship Diagram) that shows data entities that have been identified in part a) and how they relate to one another (***must be developed in LucidChart or Draw.io using crow's foot notation***). The ERD should support the transactions that have been identified in part a) and should be fully normalised to 3rd normal form (3NF).

c) List business rules for the system (at least 4 must be implemented using Transact SQL code in the CA part B). Create a data dictionary for the entity relationship diagram using the entities identified in b). The data dictionary should identify the tables, attributes, primary and foreign keys, data types and any constraints/ business rules from your chosen organisation.

d) Individual contribution report.

Formatting instructions:

Font: Cailbri ; Font Size: 12; Line Spacing: 1.15

***NO PDF Files and NO Compressed Files** – must be an MS WORD document.*

Group Management

Refer to the College's Group Work Policy, QAH B.5.2.2.2.

Group Sizes:	3
Group allocation:	Students choose
Grading approach:	Single Grade for whole group (+-10% indiv.)
Alternative assessment:	Learners with PMCs that require an alternative assessment must log a ticket for consideration.

Alternative Assessment

- An individual who cannot participate in a group must undertake the task… as an individual report

Dublin Business School

Rubric

		Evaluative Criteria/Performance Level Descriptors (what does performance at this level look like?)				
Scale	**Grade Band**	**Business Case Scope, Requirements & Business Rules**	**ERD (Entity Relationship Diagram) and Data Dictionary**	**Report Layout**	**Individual Contribution**	**Research effort**
Exceptional	**90-100%**	Business case and scope are outstanding; requirements and business rules are comprehensive, clear, and well-integrated. Demonstrates deep research and creativity.	Excellent ERD with 6+ entities, correct use of datatypes, PKs, FKs, and relationships; crows-foot notation is flawless.	The report is excellently structured, well-organized, and easy to navigate; all sections (e.g., conclusion, references) included. Clear flow, professional presentation.	All team members contribute fully, with detailed, well-organized work (500+ words each);	Strong evidence of research and problem-solving
Outstanding	**80-89%**	Business case and scope are very clear; requirements and business rules are well-written with few minor errors. Strong evidence of research.	Very good ERD with 5+ entities and mostly correct relationships, keys, and datatypes; crows-foot notation mostly accurate.	The report is very well organized; minor issues with headings or flow but generally clear and logical. All necessary sections included.	All team members contribute substantially, with good detail and evidence of research (500 words each).	Considerable amount of research and problem-solving.
Excellent	**70-79%**	Business case and scope are clear and well-defined; some elements of requirements or business rules are missing or incorrect.	ERD with 5 substantial entities; minor issues with datatypes, relationships, or keys, but largely correct.	Report layout is good; clear structure but missing some sections or minor formatting issues.	All team members contribute adequately (500 words or slightly less), showing some evidence of research and planning.	Good amount of research and problem-solving
Very Good	**60-69%**	The business case and scope are clear, but there are	ERD with 4 entities, but several minor errors with	The report is well laid out but contains some structural or	Contributions are provided by most team members, but some are	Some amount of research and problem-solving.

		some errors or missing elements in the requirements or business rules.	datatypes, PKs, or FKs; crows-foot notation has minor issues.	formatting issues; some sections may be unclear or missing.	less detailed or under 500 words.	
Good	**50-59%**	Business case and scope are present, but the requirements and business rules are incomplete or somewhat unclear.	ERD with 4 entities, some issues with datatypes, keys, or relationships, and/or incomplete crows-foot notation.	The report is generally clear but has significant formatting or organizational issues; some sections are unclear.	Contributions are minimal or some members provide less than 400 words.	Evidence of research is weak.
Sufficient	**40-49%**	Some or all elements of the business case and scope are poorly defined; major issues with requirements or business rules.	Only 3-4 entities; serious issues with datatypes, keys, relationships, or crows-foot notation.	Report layout Is disorganized, missing several important sections; difficult to follow or incomplete.	Some members fail to contribute or contribute very little; less than 400 words and lack detail.	Evidence of research is weak and lacks details.
Somewhat fulfilled	**30-39%**	Very poor or incomplete business case and scope; many missing or incorrect requirements or business rules.	Less than 3 entities or missing ERD; major issues with datatypes, keys, and relationships.	The report is poorly organized, many sections missing or confused; major issues with formatting.	No contributions or significant lack of contributions by one or more members.	Basic evidence of research is weak.
Limited fulfilment	**20-29%**	Business case and scope are missing or very poorly done; little or no requirements or business rules.	No ERD or highly inaccurate schema with no valid relationships or keys.	No report or completely disorganized, lacking necessary sections and structure.	No contribution from team members or contributions are completely inadequate.	No significant research done.

General Assessment Submission Requirements for Students:

1. Online assignments must be submitted no later than the stated deadline.

2. All relevant provisions of the Assessment Regulations must be complied with.

3. Extensions to assignment submission deadlines will not be granted, other than in exceptional circumstances. To apply for an extension please log a service desk ticket https://servicedesk.dbs.ie/HEAT . Please supply a medical cert or other required documentation.

4. Students are required to retain a copy of each assignment submitted, and the submission receipt.

5. There is a penalty of 2 marks per day for late submissions.

6. Assignments that exceed the word count will be penalised.

7. Students are required to refer to the assessment regulations in their Student Guides and on the Student Website.

8. Dublin Business School penalises students who engage in academic impropriety (i.e., plagiarism, collusion and/or copying). Please refer to the attached referencing guidelines for information on correct referencing.

What is referencing and why is it necessary?

Please follow this link to the Harvard Style Referencing Guide - all referencing is required in this format.
https://libguides.dbs.ie/skills/referencing/harvard